

Collin Bolles

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Education

Boston University

M.S. Robotics and Autonomous Systems

Rochester Institute of Technology

B.S. Computer Science

Employment

Software and Application Innovation Lab

Senior Fullstack Software Engineer

Nov. 2025 to Present

Fullstack Software Engineer

May 2022 to Nov. 2025

- Developed custom fullstack applications for R1 researchers
- Handled migration of 20+ projects from Red Hat OpenShift (K8s) to AWS
- Scoped, owned, wrote, and deployed multiple fullstack applications from start to finish
- Tools and Technology: AWS, Kubernetes, OpenShift, Typescript, React, NestJS, MongoDB, PostgreSQL

D3 Engineering · Embedded Engineering Technician

Jan. 2021 to Dec. 2021

- Developed hardware drivers for embedded computer vision applications
- Customized NVIDIA/Linux Kernel to expand vision capabilities
- Debugged newly developed hardware systems
- Tools and Technology: C, Python, Linux Kernel, Git, Gerrit, Logic Analyzer

Herrick Technology Labs · Software Intern

Sep. 2020 to Dec. 2020

- Worked on an inter-disciplinary team tasked with producing a multi-faceted sensing solution for US military avionics
- Wrote libraries to communicate with HTL radio solutions over a range of protocols including TCP over Ethernet and UART
- Tools and Technology: C++, Python, Git, Cuda

Blue Spiral · Software Developer

June 2020 to Aug. 2020

- Wrote iOS application for on-the-spot employee performance reviews using SwiftUI
- Modernized OCR system using Azure OCR API
- Leveraged satellite imagery and computer vision techniques for monitoring vegetation health
- Tools and Technology: C#, Swift, Python

Ball Bowler · Software Developer

Jan. 2019 to Nov. 2019

- Designed and developed a UI for a miniature bowling lane
- Captured state of bowling pins using computer vision
- Tools and Technology: Java, JavaFX

Blue Spiral · Software Intern

Aug. 2017 to Aug. 2018

- Developed object detection training pipeline built on top of Tensorflow
- Applied object detection pipeline for the detection of unwanted ducks
- Incorporated object detection pipeline with the Microsoft HoloLens
- Tools and Technology: C#, Python, C++, Tensorflow

Valogix · Software Intern

July 2016 to Nov. 2018

- Created a platform for real time monitoring of 100+ customer VMs
- Resolved bugs and incorporated features in the existing Valogix code base
- Tools and Technology: Java, Spring, Spring Boot, PostgreSQL, Groovy

Activities

Electric Vehicle Team

Firmware Lead

May 2020 to May 2022

Firmware Member

Dec. 2018 to May 2020

- Firmware development for custom built electric motorcycles
- Created firmware for a high-demand battery management system
- Produced a highly adaptable software interface layer for the STM32 platform
- Developed drivers across communication protocols (CAN, UART, I2C, PWM)
- Unified the inter-board communication using CANOpen